

Project Initialization and Planning Phase

Date	8 July 2026
Team ID	SWETID-2026-5556
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report:

The proposal report aims to transform credit card approval using machine learning, boosting operational efficiency and assessment accuracy. It tackles current system inefficiencies, promising faster processing, reduced credit default risks, and improved applicant satisfaction. Key features include an advanced machine learning-based credit risk model and an interactive dashboard for real-time decision-making.

Project Overview	
Objective	The primary objective is to revolutionize the credit card onboarding process by implementing advanced machine learning techniques, ensuring faster and more accurate credit risk assessments.
Scope	The project comprehensively assesses and enhances the credit card validation workflow, incorporating predictive machine learning models for a more robust, fraud-resistant, and efficient evaluation system.
Problem Statement	
Description	Addressing manual data entry inaccuracies and operational inefficiencies in the current credit card approval workflow, which adversely affects bank processing speeds and customer onboarding experiences.
Impact	Solving these issues will result in improved operational efficiency, significantly reduced credit default risks, an overall enhancement of the automated underwriting process, and a boost to organizational success.
Proposed Solution	
Approach	Employing machine learning classification techniques to analyse and predict applicant creditworthiness, creating a dynamic, data-driven, and highly adaptable credit card approval system.
Key Features	• Implementation of a machine learning-based credit scoring assessment model. • Real-time decision-making engine for instantaneous card application approvals. • Continuous model learning loops to adapt to evolving spending and financial landscapes.

Resource Requirement Table

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU (or standard multi-core CPU)
Memory	RAM specifications 8 GB / 16 GB RAM	8 GB / 16 GB RAM
Storage	Disk space for data, models, and logs	512 GB / 1 TB SSD
Software		
Frameworks	Python web/application frameworks	Flask / Django
Libraries	Data science and ML libraries	scikit-learn, pandas, NumPy, matplotlib, seaborn
Development Environment	Integrated Development Environment (IDE)	Jupyter Notebook, PyCharm / VS Code
Data		
Data	Source, size, format	Credit Card Approval Dataset (Kaggle/UCI repository), .csv format.